

# Exercise Sheet 9

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## Diffusion and Epidemics

5942UE Network Science

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# 9.A Diffusion as a Network Process

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## Exercise 9.A.1: Simple Diffusion Trace

Directed graph:  $A \rightarrow B, A \rightarrow C, B \rightarrow D, C \rightarrow D, D \rightarrow E$ .

Node  $A$  is active at  $t = 0$ . At each step, an active node activates all out-neighbours.

1. Trace the active set at  $t = 1, 2, 3$ .
2. Which node is never reached? Why?
3. If you add the edge  $B \rightarrow C$ , does anything change for this seed?

## Solution:

1.  $t = 0: A$ .  $t = 1: A, B, C$ .  $t = 2: A, B, C, D$ .  $t = 3: A, B, C, D, E$ .
2. All nodes are reachable from  $A$ . (An isolated node with no incoming edges from this set would never be reached.)
3. Adding  $B \rightarrow C$  changes nothing —  $C$  is already activated at  $t = 1$  via  $A \rightarrow C$ . Redundant paths don't accelerate spread beyond what existing paths provide.

# Exercise 9.A.2: Structural Bottlenecks

Two 8-node graphs with seed  $S$ :

- **Star:**  $S$  connected to every other node.
  - **Line:**  $S - A - B - C - D - E - F - G$ .
1. How many steps until all nodes are active in each?
  2. Where is the structural bottleneck?
  3. A single edge fails. Which topology is more affected?

## Solution:

1. Star: 1 step. Line: 7 steps.
2. The line has a bottleneck at every edge — removing any node severs everything downstream. The star has no internal bottleneck.
3. Removing one edge from the line blocks up to 6 of 7 nodes. Removing one edge from the star isolates exactly one node. The line is vastly more vulnerable.

# Exercise 9.A.3: Centrality and Diffusion Speed

Load the karate club graph and BFS-diffuse from the highest-degree node.

1. How many steps until all 34 nodes are active?
2. Repeat from the lowest-degree node. How does the curve change?
3. What does this say about hubs?

## Solution:

```
import networkx as nx

G = nx.karate_club_graph()
cent = nx.degree_centrality(G)
seed = max(cent, key=cent.get)

active, frontier = {seed}, {seed}
for t in range(8):
    new_frontier = {nbr for node in frontier for nbr in G.neighbors(node)
                    if nbr not in active}
    active |= new_frontier
    frontier = new_frontier
    print(f"t={t+1}: {len(active)} active")
```

A hub seed reaches the full graph in 3–4 steps. The lowest-degree node takes 5–6+ steps with a much flatter early curve. Superspreaders have outsized impact because they bridge to the rest of the network in fewer hops.

## Exercise 9.A.4: Paths and Reachability

Company-wide email cascade from CEO ( $C$ ) must reach 500 employees. The network has a core-periphery structure: 50 managers in a dense core, each linked to  $\approx 10$  peripheral employees.

1. Does a message from  $C$  to a peripheral employee require passing through the core?
2. If 5 of 50 managers are absent, what fraction of periphery is potentially unreachable?
3. How could the company become more resilient?

## Solution:

1. Yes — peripheral nodes only link to core nodes, so every path passes through the core.
2. Each absent manager isolates  $\approx 10$  peripheral nodes  $\rightarrow$  about  $50/450 \approx 11$  unreachable if no redundant paths exist.
3. Add cross-edges between peripheral employees; give each peripheral employee two or more core contacts; identify high-betweenness bridge managers and ensure backup coverage. Resilience requires *redundant paths*.

# 9.B Epidemic Models and Simple Contagion

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## Exercise 9.B.1: SIR Trace

SIR rules (synchronous):  $S \rightarrow I$  if any infected neighbour, with probability  $\beta$ .  $I \rightarrow R$  at the next step (single-step infectious period).

Path graph  $A - B - C - D - E$  with  $B$  infected at  $t = 0$ ,  $\beta = 1.0$ .

1. Trace each node's state at  $t = 0, 1, 2, 3, 4$ .
2. When does  $E$  become infected?
3. With  $\beta = 0.5$ , what is  $P(C \text{ not infected at } t = 1)$ ?

## Solution:

### 1. State trace:

| Time    | A | B | C | D | E |
|---------|---|---|---|---|---|
| $t = 0$ | S | I | S | S | S |
| $t = 1$ | I | R | I | S | S |
| $t = 2$ | R | R | R | I | S |
| $t = 3$ | R | R | R | R | I |
| $t = 4$ | R | R | R | R | R |

2.  $E$  is infected at  $t = 3$ , recovers at  $t = 4$ .

3.  $C$  has one infected neighbour ( $B$ ), so it is infected with probability 0.5.  $P(\text{not infected}) = 0.5$ .

# Exercise 9.B.2: Structural Effect of Topology

SIR with  $\beta = 1.0$ , one-step recovery, seed = node 1.

- **Line L:** 1 — 2 — 3 — 4 — 5 — 6
  - **Star S:** node 1 is the hub of 2, 3, 4, 5, 6
1. Maximum simultaneously infected in L?
  2. Maximum simultaneously infected in S?
  3. Which produces a sharper epidemic peak? Implications?

## Solution:

1. Line: at any time only 1 node is infected (a moving wave).
2. Star: at  $t = 1$ , all 5 spokes are infected simultaneously.
3. The star produces a sharper, taller peak. **High-degree hubs create epidemic explosions** — a single infected hub exposes all neighbours at once. This is why superspreader events are dangerous and why targeted vaccination of hubs is so effective.

# Exercise 9.B.3: SIR Simulation in NetworkX

Implement synchronous SIR on the karate club graph and analyse it.

1. At what step does the epidemic peak?
2. What fraction ends in  $R$  (ever infected)?
3. Vary  $\beta$  from 0.1 to 0.8. Where does the epidemic reliably infect  $> 80$ ?

## Solution:

```
import networkx as nx
import random

def sir_simulation(G, seed_node, beta=0.3, steps=20):
    state = {n: 'S' for n in G.nodes()}
    state[seed_node] = 'I'
    history = []
    for _ in range(steps):
        new_state = state.copy()
        for node in G.nodes():
            if state[node] == 'I':
                new_state[node] = 'R'
            elif state[node] == 'S':
                inf_nbrs = [n for n in G.neighbors(node) if state[n] == 'I']
                if any(random.random() < beta for _ in inf_nbrs):
                    new_state[node] = 'I'
        state = new_state
        history.append(tuple(sum(1 for s in state.values() if s == k)
                                for k in ('S', 'I', 'R')))
```

With  $\beta = 0.3$  and a hub seed, the peak hits around step 3–5. Below  $\beta \approx 0.1$  the epidemic dies out; above  $\beta \approx 0.4$  nearly the whole network is eventually infected. The dense community structure makes the karate club highly susceptible once the threshold  $R_0 > 1$  is crossed.

## Exercise 9.B.4: Simple vs. Complex Contagion

For each scenario, decide whether it is **simple** (one contact sufficient) or **complex** (multiple confirmations needed):

(a) A cold virus. (b) Adopting a new software tool only after three colleagues recommend it. (c) Sharing a news article posted by one friend. (d) Joining a protest after pressure from multiple community members. (e) Updating a belief after seeing a claim from several independent sources.

For (b) and (d), how must network structure differ from simple contagion?

## Solution:

| Scenario     | Type    |
|--------------|---------|
| (a) Cold     | Simple  |
| (b) Software | Complex |
| (c) Article  | Simple  |
| (d) Protest  | Complex |
| (e) Belief   | Complex |

For (b) and (d) to spread, the network needs **dense local clusters** where multiple neighbours have already adopted. A bridge or weak tie — which excels for simple contagion — fails for complex contagion because the node on the other side has only one adopter neighbour.

**Complex contagion spreads through tight communities and stalls at bridges** — exactly the opposite of simple contagion, which exploits bridges for rapid long-range diffusion.

# Wrap-Up

- diffusion is bounded by reachable paths; redundant paths don't help unless primary paths fail
- bottlenecks and hubs control epidemic dynamics
- the same  $\beta$  produces dramatically different curves on different topologies
- complex contagion reverses the role of bridges and clusters compared to simple contagion